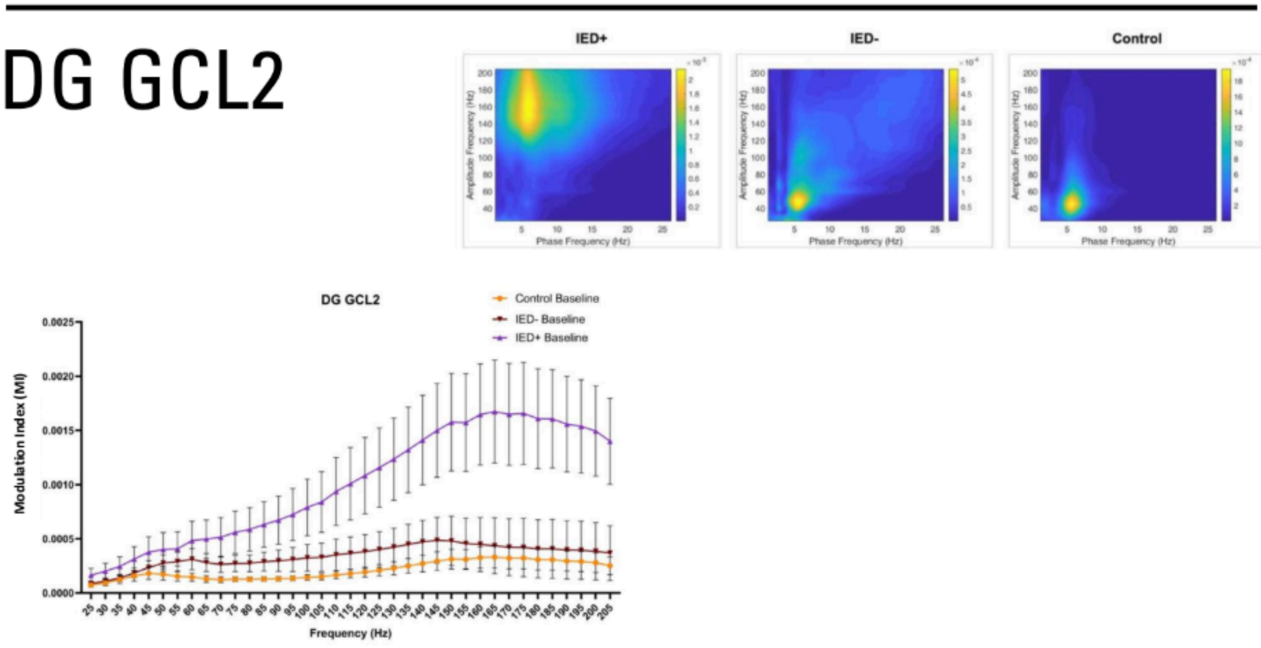
Dear ShahriarGPT,



Hey there. So last spring, for my thesis, we included these. On the top right of this image we have our commodulograms (Phase frequency of theta on the x and amplitude frequency of gamma on the y) and on the bottom of this image is a general estimating equations (GEE) model, showing the correlation between gamma frequency (Hz) on the x axis, and modulation index of theta (4-12 Hz) relative to these gamma frequencies. The commodulograms are how we derive modulation index. Where you see the warmer color, the higher the modulation index. These graphs correlate to each other as we derive our GEE model from the commodulograms.

The data is nice here, but averaging it across a whole recording takes away key things that are occurring in the background. We don't at what time the main effects are occurring, we don't know what theta Hz is driving it, and how/if theta power is influencing our results.

So… to combat these issues we are planning on taking on a different approach.

To get at this time issue we are going to create epochs of 6 seconds. This will give us time resolution so that we can actually see variance.

An example of the graph is here:

G
a
m
m
a

(y)
Theta (x)

Epoch 1: 0-6 sec

And this epoch will be a M.I. array.

We will have these for the whole recording. PER CHANNEL. So there will be about 63.

G
a
m
m
a

(y)
Theta (x)

Epoch 1: 0-6 sec

G
a
m
m
a

(y)
Theta (x)

Epoch 2: 6-12 sec

G
a
m
m
a

(y)

Theta (x)

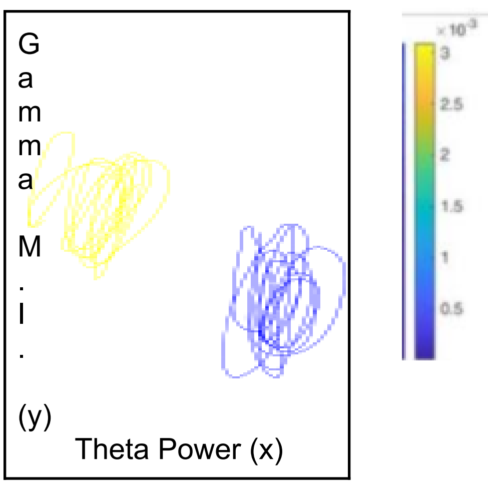
Epoch 3: 12-18 sec

At the same time, we want to get the theta power for the epoch…. For every epoch we have a theta power for 4hz, 5hz, 6hz, 7hz, 8hz…12hz. We want the x values for these theta power frequencies.

So the end result of this first step is a MI 3D array… of Theta x Gamma x Epoch.
And the theta power 2D array… Theta power x Epoch

This ultimately leads us to… how does the power of theta change the M.I.?

At 4 Hz…



And this is a heat map… this will also be by channel…. So we can really get what channels correlate to specific layers down.